

Cooling of non-ideal gases

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For real gases, the Mendeleev-Clapeyron equation $pV = \nu RT$ is satisfied approximately and the more accurately, the lower the density of the gas and the higher its temperature. In real gases, forces act between molecules, and at close distances these are repulsive forces, and at large distances they are attractive. One of the simple equations that takes into account the interaction of molecules is the Van der Waals equation of state:

$$\left(p + \frac{a\nu^2}{V^2}\right)(V - \nu b) = RT$$

Где a и b — положительные коэффициенты Ван-дер-Ваальса, которые в этой задаче можно считать постоянными. Член $\frac{a\nu^2}{V^2}$ описывает уменьшение давления газа из-за взаимного притяжения молекул, а νb - takes into account the finite size of the molecules. Note that the van der Waals equation of state can qualitatively describe both the behavior of a gas and the behavior of a liquid.

The Van der Waals equation of state is a third degree equation in coordinates (p, V) . In Fig. Figure 1 shows a series of isotherms of one mole of van der Waals gas. The real isotherm differs from the van der Waals isotherm by a straight section AB with constant pressure p_{LG} , located along the volume axis between V_L and V_G , on which the equilibrium of liquid (indicated by the index L) and gas (indicated by the index G) is realized. Using the second law of thermodynamics, J. Maxwell showed that the pressure p_{LG} should be chosen in such a way that those shown in Fig. 2 areas I and II were the same. Also, the internal energy of a van der Waals gas depends not only on temperature, but also on volume:

$$U = \nu C_V T - \frac{a\nu^2}{V}$$

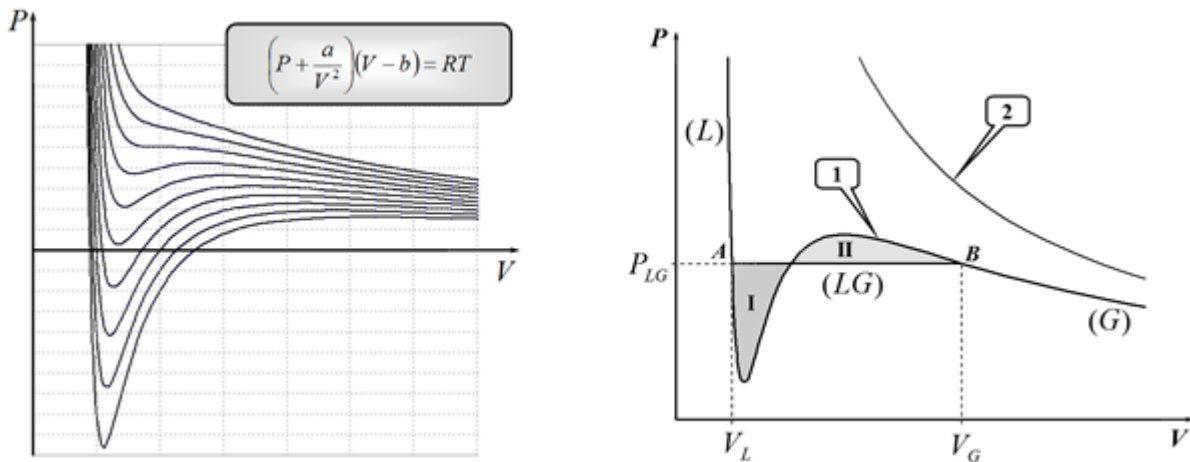


Fig. 1.

Примечание: one mole of substance is used in all parts of the problem.

A1 One of the industrial methods of cooling (and subsequent liquefaction) is the adiabatic and quasi-static expansion of gases. Let the gas expand from the initial volume V_1 to the final volume V_2 . For a gas that obeys the van der Waals equation of state, in this process the temperature drops by ΔT_W , and for an ideal gas - ΔT_I . The heat capacity C_V of a van der Waals gas is taken equal to the heat capacity of an ideal gas C_V at constant volume. Find the ratio $\Delta T_W/\Delta T_I$. Express your answer in terms of V_1 , V_2 , a , b , C_V and R . In which case is there more cooling? **0.70**

A2 The gas can also be cooled by allowing it to expand adiabatically into a vacuum. Let the gas be in the vessel at the initial moment, occupying volume V_1 in it. After removing the partition, the gas is able to freely expand to volume V_2 . Assuming the vessel is thermally insulated, find the temperature change for the cases of van der Waals gas ΔT_W and ideal gas ΔT_I . Express your answer in terms of V_1 , V_2 , a , b , C_V and R . In which case is there more cooling? **0.40**

A3 Gas can be converted into a liquid state by compressing it isothermally. However, to do this, the gas must first be cooled below a certain temperature T_C . Above this temperature, gas cannot be converted into liquid. Using the van der Waals equation of state, find this temperature T_C . Express your answer in terms of a , b and R . **0.90**

In 1852, the effect was discovered, which was subsequently widely used in industry for cooling and liquefying gases. The Joule-Thompson effect is the change in gas temperature during adiabatic throttling - the slow flow of gas under the influence of a constant pressure difference through a porous partition (throttle), as shown in Fig. 3. To the left of the throttle there is gas at pressure p_1 (it is maintained constant), after gas flows through the throttle the gas pressure becomes equal to p_2

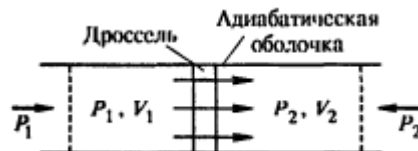


Fig. 2.

B1 Let us consider a portion of gas with volume V_1 located to the left of the throttle. Having passed through the throttle, this portion of gas occupies volume V_2 . Using the first law of thermodynamics, show that the function $W = U + pV$ (enthalpy) is conserved in the process under consideration, that is, its values are the same for the same portion of gas to the right and left of the partition. Assume that there is no heat exchange with the environment. **0.70**

B2 Let the Joule-Thompson process use an ideal gas. Find the decrease in its temperature ΔT_I . The heat capacity of gas is C_V . **0.30**

In paragraphs C of the problem the so-called integral Joule-Thompson effect is considered. This is an effect that occurs with large pressure differences on different sides of the throttle. As a model of the integral effect, you can use the following: to the left of the throttle, the gas is at high pressure and is described by the van der Waals equation of state, and after passing through the throttle, the gas can be considered ideal.

C1	Find the decrease in gas temperature ΔT_W in such a process. The heat capacity of gas is C_V . Express your answer in terms of a, b, R, C_V, T_1 and V_1 .	1.00
C2	If the initial state of the gas is depicted on a diagram (T_1, V_1) , then a curve can be drawn on it that divides the plane into two regions: the points of one region correspond to $\Delta T_W < 0$ (the gas is cooled), and the other $\Delta T_W > 0$ (the gas is heated). This curve is called the inversion curve of the integral Joule-Thompson effect. Find the equation of this curve, i.e. express T_{inv}^l through a, b, R, C_V and V_1 . Draw the curve on a graph and indicate the characteristic features of the curve. On the graph, indicate the areas where the gas cools and warms up.	0.80
C3	However, it is more convenient to draw the inversion curve of the Joule-Thompson effect in coordinates (p_1, T_1) , because it is these gas parameters that can be directly measured. Find the equation of this curve, i.e. express T_{inv}^l in terms of a, b, R, C_V and p_1 (or p_1 in terms of a, b, R, C_V and T_{inv}^l). Draw the curve on a graph and indicate the characteristic features of the curve. On the graph, indicate the areas where the gas cools and warms up.	1.00

Part D of the problem considers the Joule-Thompson effect in the general case. In this case, the gas on both sides of the partition is described by the van der Waals equation of state. If the gas to the left of the partition is at an inversion temperature, then during throttling the gas temperature does not change.

D1	Derive an expression for the inversion temperature T_{inv} of the Joule-Thompson effect in the general case. Express it using a, b, R, C_V, V_1 and V_2 .	1.00
D2	The differential Joule-Thompson effect is realized at small pressure drops. For it we can assume that $V_1 \approx V_2$. Under this condition, obtain an expression for the inversion temperature T_{inv}^D of the differential Joule-Thompson effect. Express it using a, b, R, C_V and V_1 .	0.20
D3	Find the equation of the inversion curve of the differential Joule-Thompson effect, i.e. express T_{inv}^D in terms of a, b, R, C_V and p_1 (or p_1 in terms of a, b, R, C_V and T_{inv}^D). Draw a schematic diagram of the curve on the graph (p_1, T_1) , indicate the characteristic features of the curve. On the graph, indicate the areas where the gas cools and warms up.	1.50

Part E examines the relationship between the integral and differential Joule-Thompson effects.

E1	Sketch the inversion curves for both effects on the same graph in coordinates (p_1, T_1) . Identify the areas where the gas is cooled and heated for both effects. Explain the existence of regions where the temperature change has a different sign for the integral and differential effect.	0.50
E2	For the industrial use of the integral Joule-Thompson effect for a certain temperature T_1 , it is important to select such a pressure p_1 so that the cooling of the gas in the process is maximum. These relationships can also be represented on a graph (p_1, T_1) . Draw this relationship diagrammatically. On the same graph, schematically depict the inversion curves for the differential and integral effects on the same graph.	1.00

Source: <https://pho.rs/p/300>